Assignment 4

link : https://www.notion.so/Assignment-4-1fdee40c6a08804cb260f8a75aece5e4?pvs=4

1. How many times will the following code print “Welcome to Java”?

   ```
   10
   ```

2. What will be displayed when the following code is executed?

   ```
   3 0
   ```

3. What is the value in count after the following loop is executed?

   ```
   10
   ```

4. The following loop displays

   ```
   1 3 5 7 9
   ```

5. Analyze the following fragment:

   Answer: The program will never stop because d will never equal to 10.0, because double number 0.1 is not exactly equal to 0.1, it is 0.099999.

6. How many times is the println statement executed?

   ```
   45
   ```

7. Suppose the input is 2 3 4 5 0. What is the output of the following code?

   ```
   max is 5
   number is 0
   ```

8. What are the differences between a while loop and a do-while loop? Convert the following while loop into a do-while loop

   Answer: while loop execute when the condition is true, do-while loop execute at least once before the condition.

```
int sum, number;
do {
    System.out.println("Enter an integer " + ".....");
    number = input.nextInt();
    sum += number;
    } while (number != 0)
```

9. The following loop displays:

```
      1
     2 1
    3 2 1
   4 3 2 1
  5 4 3 2 1
6 5 4 3 2 1
```

10. What is the keyword break for? What is the keyword continue for? Will the following programs terminate? If so, give the output.

    Answer: break will stop the loop, continue will skip the current loop and go into next loop.

    i.

```
Balance is 1
```

    ii.

```
infinite loop after balance <9
```

11. The for loop in (i) is converted into the while loop in (ii). What is wrong? Correct it.

```
if (i % 3 == 0) {
    i++;
    continue;
}
```

12. Rewrite the code segment below without using continue.

```
int sum = 0, number = 0;
while (number < 20) {
    number++;
    if (number != 10 && number != 11)
        sum += number;
    System.out.println("The sum is " + sum);
```

13. Produce problem analysis chart, flowchart and Java program for the following problem.

Data	Processing	Output
- numberOfStudents - studentName - numberOfSubjects - totalScore	Read numberOfStudents While i ≤ numberOfStudents Read studentName, numberOfSubjects, totalScore Calculate TPScore = totalScore/numberOfSubject Print studentTP and studentTPGrade Print totalStudentNumberOfEachGrade	- studentTP - studentTPGrade - totalStudentNumberOfEachGrade

```
import java.util.Scanner;

public class Test {
    public static void main(String[] args) {
```

```
        Scanner input = new Scanner(System.in);
        int numberOfSubjects, totalScore;
        String studentName, TPGrade;
        double TPScore;
        int countTP5 = 0 , countTP4 = 0, countTP3 = 0, countTP2 = 0;
        int countTP1 = 0, i = 0;

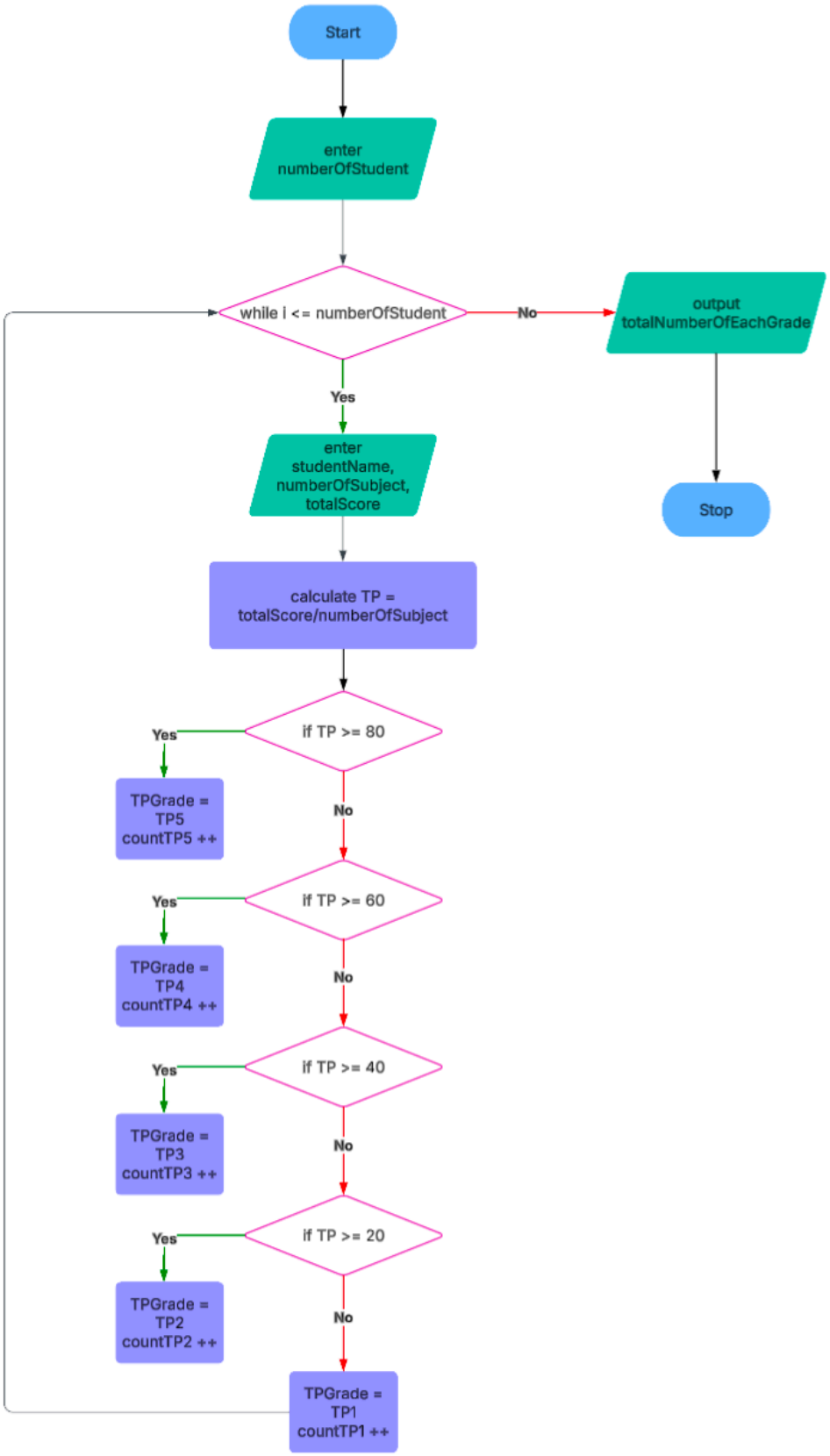
        System.out.println("TP Calculation");
        System.out.print("Number of students: ");
        int numberOfStudents = input.nextInt();
        while (i < numberOfStudents) {
            System.out.print("Insert your name: ");
            studentName = input.next();
            System.out.print("Total subject: ");
            numberOfSubjects = input.nextInt();
            System.out.print("Total subject score: ");
            totalScore = input.nextInt();
            TPScore = (double)totalScore / numberOfSubjects;
            if (TPScore >= 80) {
                TPGrade = "TP5";
                countTP5++;
            } else if (TPScore >= 60) {
                TPGrade = "TP4";
                countTP4++;
            } else if (TPScore >= 40) {
                TPGrade = "TP3";
                countTP3++;
            } else if (TPScore >= 20) {
                TPGrade = "TP2";
                countTP2++;
            } else {
                TPGrade = "TP1";
                countTP1++;
            }
            System.out.println("Student TP is " + TPScore + " - " + TPGrade);
            System.out.println("....\n");
```

```
            i++;
        }

        System.out.println("The number of students based on TP category");
        System.out.println("TP5 - " + countTP5);
        System.out.println("TP4 - " + countTP4);
        System.out.println("TP3 - " + countTP3);
        System.out.println("TP2 - " + countTP2);
        System.out.println("TP1 - " + countTP1);

    }
}
```

Note: should have add condition statement to check if all inputs are valid, but this is enough to work in normal condition.


Start
enter numberOfStudent
while i <= numberOfStudent
No
output totalNumberOfEachGrade
Stop
Yes
enter studentName, numberOfSubject, totalScore
calculate TP = totalScore/numberOfSubject
if TP >= 80
Yes
TPGrade = TP5 countTP5 ++
No
if TP >= 60
Yes
TPGrade = TP4 countTP4 ++
No
if TP >= 40
Yes
TPGrade = TP3 countTP3 ++
No
if TP >= 20
Yes
TPGrade = TP2 countTP2 ++
No
TPGrade = TP1 countTP1 ++